

DEVELOPER INFRASTRUCTURE PITCH DECK

rapid_ingestor

Kernel-Bypass, Zero-Allocation Telemetry & Log
Ingestion Engine

Seastar C++ Reactor Framework

10x Vector Throughput



Traditional Telemetry Middleware Hits the POSIX & Kernel Wall

Traditional Middleware (Vector / FluentBit)

- ✗ **Kernel Context Switching:** Millions of POSIX socket syscalls daily
- ✗ **Thread Lock Contention:** epoll and OS thread scheduling overhead
- ✗ **Disk Flushing Stalls:** OS page cache locks during burst writes
- ✗ **Write Amplification:** Direct unbatched raw logs choke ClickHouse

The rapid_ingestor Paradigm

- ✓ **User-Space Networking:** Zero kernel context switches on packet ingress
- ✓ **Thread-Per-Core Locks:** Isolated core execution without lock sharing
- ✓ **Direct DMA Disk Access:** Direct io_uring and O_DIRECT bypassing page cache
- ✓ **Native Columnar Formatting:** Zero fragmentation writes for ClickHouse

End-to-End Kernel Bypass & Zero-Allocation Pipeline



Kernel-Bypass Networking

DPDK and AF_XDP ring buffer ingress directly to user memory.



Thread-Per-Core Reactor

Share-nothing CPU pinning eliminating thread contention locks.



Direct Async Disk I/O

Linux io_uring and O_DIRECT bypassing OS page cache stalls.

EMPIRICAL PERFORMANCE PROOF

10x Throughput Advantage Over Vector

1.16M

Messages / sec steady ingestion throughput

>10x

Outperformance vs Vector rust runtime

237 MB/s

Deterministic flat-line wire bandwidth

Benchmarked on single core socket parsing with 204.7-byte average log payloads.

Stream-to-Storage Processing Chain

STEP 01

Ring Ingress

Zero-copy kernel bypass packet capture.



STEP 02

Zero-Alloc Parse

Native protocol extraction in user memory.



STEP 03

Column Batch

Direct ClickHouse native block formatting.



STEP 04

Direct Commit

io_uring flush without page cache locks.

Open-Core Engine to Enterprise Control Plane

1. Open-Source Core

High-performance C++ ingestion engine distributed freely to build developer adoption.

Bottom-Up Dev Adoption

2. Enterprise Plugins

Monetize corporate identity extensions, LDAP input controls, and compliance tools.

Inbound Enterprise Pull

3. Managed Cloud

Hosted control plane managing multi-cluster ClickHouse and analytical pipelines.

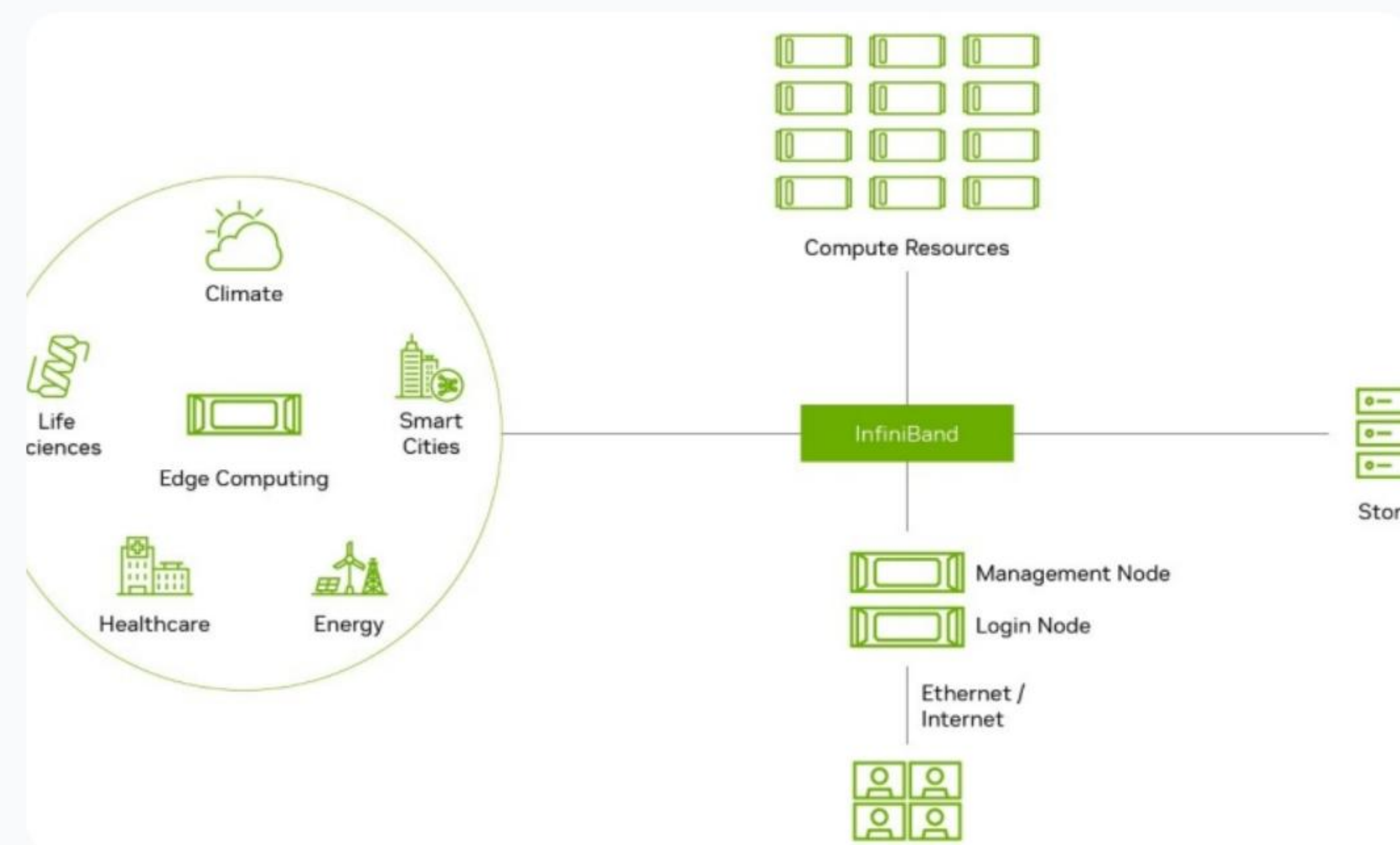
SaaS Control Plane

3 Decades of Low-Level Systems Mastery

Solo Technical Founder

Built, optimized, and benchmarked rapid_ingestor independently.

- </> Deep Systems Expertise:** Extensive track record in C++, Seastar, DPDK, AF_XDP, eBPF, and Go.
- 🗄️ Datastore Mastery:** Direct experience optimizing ClickHouse and MongoDB storage boundaries.
- ⚡ Execution Power:** Demonstrated ability to single-handedly engineer breakthrough infrastructure.



HEAVYBIT PARTNER INVESTMENT ASK

Building the Next-Gen Telemetry Standard

Raising Day 0 / Pre-Seed capital to convert early inbound pull into enterprise design partnerships and scale core engineering.

MILESTONE 1

Design Partners

Deliver LDAP enterprise inputs for initial design cohorts.

MILESTONE 2

ClickHouse Ecosystem

Release native zero-copy ClickHouse block format plugins.

MILESTONE 3

Team Expansion

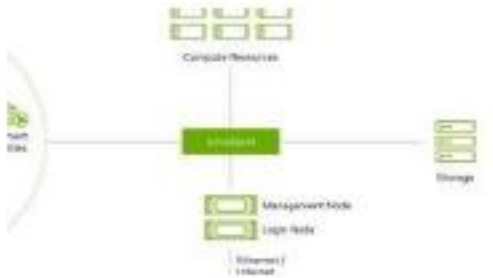
Recruit lead core systems engineer and GTM partner.

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